Band-1 Hadamard Order 6884 — Miyamoto (Double Paley-I H_1720 + C_1722) + Paley-I (q=6883) IN SPK; Gap 2052 Still OPEN

H_860 Paley-I PASS; H_1720 Double PASS; C_1722 PASS; H_6884 Miyamoto PASS; H_6884 Paley-I PASS → IN SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6836_Miyamoto`; `2026-09-15_Band1_Hadamard_6724_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6884_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 860, 1720, and 6884 only** (verified $\max|HH^\top - nI| = 0$), via (i) Miyamoto with Double Paley-I seed H_{1720} = Sylvester-double(H_{860} = Paley-I($q = 859$)) and symmetric Paley conference C_{1722} over prime $q = 1721 \equiv 1 \pmod 4$, and (ii) classical Paley-I $q = 6883$ (prime $\equiv 3 \pmod 4$) → H_{6884}. Classical SPK: Paley-I **PASS**es ($q = 6883$); Paley-II **FAIL**s ($q = 3441$ not prime); Kronecker **FAIL**s (4×1721). Therefore order **6884 is IN SPK via Paley-I** (Miyamoto is an additional finite construction, not the SPK reason). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **6836 Miyamoto-only (NOT in SPK)**; seed **1708** retained in SPK. Prior SPK seeds **1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/6788/6836/...** excluded; **860/1720/6884** folded into ONBOX when verified.

Abstract

Finite constructions of orders **860**, **1720**, and **6884**.

- H_{860} = Paley-I ($q = 859$) — Double half
- H_{1720} = Sylvester-double(H_{860}) — Miyamoto seed
- C_{1722} = Paley conference over prime $q = 1721$
- H_{6884} = Miyamoto(H_{1720}, C_{1722})
- H_{6884} = Paley-I ($q = 6883$) — classical SPK
- Paley-II for 6884: **SKIP** ($q = 3441$ not prime)

SPK: 6884 IN SPK via Paley-I q=6883. Seeds 860 / 1720 / 6884 enter SPK. Prior SPK **1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6884.py`

Results. overall_pass=true; units 99/99; wall 536.8373s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 6884 = 4 \cdot 1721$; 1721 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1720} and C_{1722} (Paley conference over prime $q = 1721$).
- Seed: Paley-I $q = 859$ (prime $\equiv 3 \pmod 4$) → H_{860} → Sylvester-double → H_{1720}.
- SPK: Paley-I $q = 6883$ **PASS**; Paley-II $q = 3441$ **FAIL**; Kronecker **FAIL** → **IN SPK via Paley-I**.
- Prior: 6836 Miyamoto-only shipped; next optional was $q = 1721 \rightarrow n = 6884$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_860 Paley-I	Paley-I q=859	0	0.6678
H_1720 Double	Sylvester-double(H_860)	0	2.3387
C_1722	Paley conf q=1721	0	3.5607
H_6884 Miyamoto	Miyamoto(H_1720 Double, C_1722)	0	250.3606
H_6884 Paley-I	Paley-I q=6883	0	273.0224
H_6884 Paley-II	SKIP q=3441 not prime	—	—
H_6884 Kronecker	SKIP $4 \cdot 1721$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **6836** lineage (**1708** Paley-II; **6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 6676/6484/6404/6388/6196/6788/6836) + verified **860** / **1720** / **6884** (Paley-I); **exclude** Miyamoto-only bonuses including **6836**. Known count 1392. Next constructive gap **2052** (OPEN).

Quantity	Value
6884 in SPK	**True** (Paley-I q=6883)
1720 in SPK	**True** (Double Paley-I)
860 in SPK	**True** (Paley-I q=859)
1708 / 840 / 1680 / 6724 in SPK	**True** (prior)
6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN.

Repro

Lab: `/workspace/experiments/band1/hadamard_6884_miyamoto/`
Artifacts: `H860_paleyI_pm1.npz`, `H1720_double_pm1.npz`, `C1722_pm1.npz`, `H6884_miyamoto_pm1.npz`, `H6884_paleyI_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.